1) **Write a ‘C’ function to check if a number is prime. Use this function to display all prime numbers between 100 and 500.**

Ans -

```
#include <stdio.h>
int isPrime(int n)
{
    int i;

    if (n < 2)
        return 0;

    for (i = 2; i <= n / 2; i++)
    {
        if (n % i == 0)
            return 0;
    }

    return 1;
}

int main()
{
    int i;

    printf("Prime numbers between 100 and 500:\n");

    for (i = 100; i <= 500; i++)
    {
        if (isPrime(i))
            printf("%d ", i);
    }

    return 0;
}
```

O/P - Prime numbers between 100 and 500:

101 103 107 109 113 127 131 137 139 149
151 157 163 167 173 179 181 191 193 197
199 211 223 227 229 233 239 241 251 257
263 269 271 277 281 283 293 307 311 313
317 331 337 347 349 353 359 367 373 379
383 389 397 401 409 419 421 431 433 439
443 449 457 461 463 467 479 487 491 499

2) Write a 'C' program to accept and display two matrices and check two matrices are identical or not.

Ans -

```
#include <stdio.h>
int main()
{
    int a[10][10], b[10][10];
    int i, j, r, c, same = 1;

    printf("Enter rows and columns: ");
    scanf("%d%d", &r, &c);

    printf("Enter first matrix:\n");
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &a[i][j]);

    printf("Enter second matrix:\n");
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &b[i][j]);

    printf("First matrix:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
            printf("%d ", a[i][j]);
        printf("\n");
    }

    printf("Second matrix:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
            printf("%d ", b[i][j]);
        printf("\n");
    }

    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            if (a[i][j] != b[i][j])
                same = 0;

    if (same == 1)
```

```
        printf("Matrices are identical");
    else
        printf("Matrices are not identical");

    return 0;
}
```

O/P -Enter rows and columns: 2 2
Enter first matrix:
1 2
3 4
Enter second matrix:
1 2
3 4

3) **Write a 'C' program to which accepts a character from the user and display its ASCII value. Also display its next and previous character.**
Ans -

```
#include <stdio.h>
int main()
{
    char ch;

    printf("Enter a character: ");
    scanf("%c", &ch);

    printf("ASCII value = %d\n", ch);
    printf("Previous character = %c\n", ch - 1);
    printf("Next character = %c\n", ch + 1);

    return 0;
}
```

O/P- Enter a character: B
ASCII value = 66
Previous character = A
Next character = C

4) **Write a 'C' program to accept a matrix of size m x n and display transpose of a given matrix.**
Ans -

```
#include <stdio.h>
int main()
```

```
{
    int a[10][10], m, n, i, j;

    printf("Enter rows and columns: ");
    scanf("%d%d", &m, &n);

    printf("Enter matrix elements:\n");
    for(i = 0; i < m; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Transpose of matrix:\n");
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < m; j++)
        {
            printf("%d ", a[j][i]);
        }
        printf("\n");
    }

    return 0;
}
```

O/P - Enter rows and columns: 2 3
Enter matrix elements:
1 2 3
4 5 6

Transpose of matrix:
1 4
2 5
3 6

5) **Write a ‘C’ program to accept real number x and integer n and calculate the sum of first n terms of the series x+ x/3!+ x/5!+ x/7!+…**

Ans - #include <stdio.h>

```
int main()
{
```

```
    float x, sum = 0, fact;
    int n, i, j;

    printf("Enter value of x: ");
    scanf("%f", &x);

    printf("Enter number of terms: ");
    scanf("%d", &n);

    sum = x;

    for(i = 1; i < n; i++)
    {
        fact = 1;
        for(j = 1; j <= 2*i+1; j++)
        {
            fact = fact * j;
        }
        sum = sum + x / fact;
    }

    printf("Sum = %f", sum);

    return 0;
}
```

O/P- Enter value of x: 2
Enter number of terms: 3
Sum = 2.400000